From Geometry to a Real-Time Digital Twin of Rolling-Disc Notched Tendon-Driven Continuum Robots

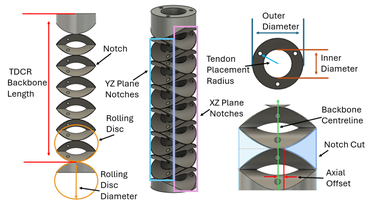
Om Mohan Pawar Supervisor: **Prof. Madhu Vadali** IITGN Robotics Laboratory, Mechanical Engineering, IIT Gandhinagar B.Tech Project, Final Evaluation, November 2026

IITGN ROBOTICS

PART A How does geometry govern compliance?

under review, ICRA 2027 ICRA IEEE INTERNATIONAL CONFERENCE ON ROBOTICS AND AUTOMATION

The robot, and two questions



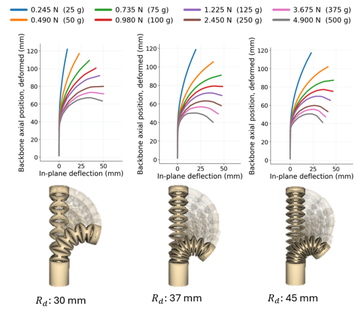
A. Design
Which geometry gives a wanted bend, and what stroke and force does it cost?

B. Model
Can the whole shape be predicted live from servo commands alone, for any stack and mounting?

TPU tube, two orthogonal families of rolling-disc notches, tendons in the wall, stackable 160 mm modules.

A1 SWEEP: geometry to simulation, unattended

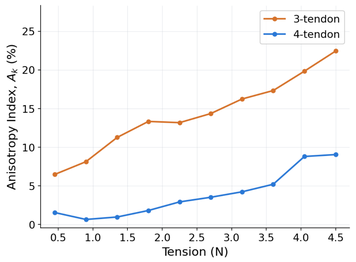
Parameters D_{RD}, L, D_o, D_i, e, c, N_t → **CAD** CadQuery, one Boolean cut → **Mesh** CGAL tetrahedra → **FEM** SOFA, pull-only tendons → **Metrics** curvature, CC error, stroke



designs	14
tendons × loads	4 × 11
FEM runs (done)	616 (614)
E, TPU, measured	56 MPa
gravity	on
claims valid for	$T \le 0.9$ N

Same load, three disc diameters: a larger disc gives more, shallower notches and a tighter bend.

A2 Match the actuation to the structure

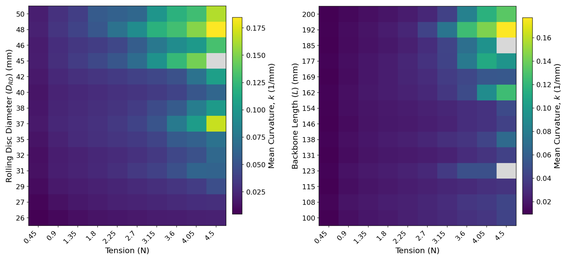


With three tendons the bend depends on its direction. The anisotropy index $A_\kappa = \sigma_\kappa / |\bar{\kappa}|$ falls from 6.2 % to 2.6 % with four tendons (13 of 14 designs).

Four tendons form two antagonistic pairs, so a section needs two motors.

Anisotropy index against tension for three and four tendons on one design.

A3 What each parameter controls



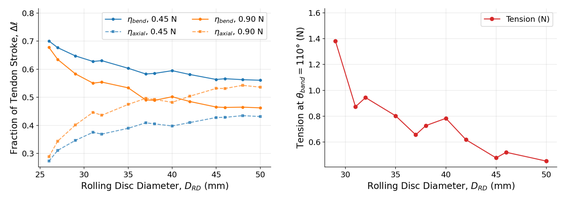
D_{RD} sets curvature turning 54° → 143° (26 to 50 mm, 0.9 N)

L sets reach tip travel 78 → 157 mm (100 to 200 mm)

Mean curvature (1/mm). More notches bend further, but deviate up to 5× more from constant curvature.

A4 Stroke budget: what a bend costs the actuator

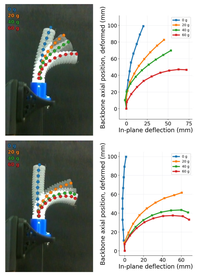
$$\underbrace{\Delta\ell}_{\text{tendon stroke}} = \underbrace{\Theta\, d}_{\text{bending}} + \underbrace{\Delta L}_{\text{axial compression}}$$



Left: fractions of the stroke spent on bending and on compression. Right: tension for a 110° bend. Both against D_{RD}.

The budget closes to 0.99 on the notched designs. Between 42 and 47 % of the stroke shortens the body instead of bending it, and a larger disc lowers the tension needed for a 110° bend from 0.87 to 0.45 N.

A5 Experiment: three printed prototypes



D_{RD} (mm)	NRMSE, % of length centreline	tip	axial
30	18.5	20.9	43.1
67.89	10.1	23.5	11.9
86.84	16.3	64.7	23.9

Shape agrees within 18.5 %; the tip less well: friction and self-contact are not modelled.

PART B A sensor-free, real-time digital twin

in preparation for IEEE RA-L IEEE Robotics & Automation Society

B1 Servo commands in, whole shape out

commands $\delta\ell$ (servos) + mounting (gravity) → **Tendon map** $\delta\ell = G\lambda$ from CAD → **Pull-only** convex QP (slack) → **Gravity** fixed point, any angle → **Shape** pose every 5 mm

No sensor on the robot. One module model, composed by statics for any stack and mounting.

B2 The physics: a beam that follows the CAD

1 a tendon loads every slice 2 it shortens and bends 3 tendons only pull

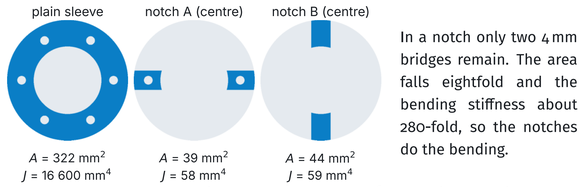


$$\delta\ell = G\,\lambda, \quad G_{ij} = \sum_c \tau_{ic}\tau_{jc}\left(a_c + \boldsymbol{d}_i^{\top} C_c\, \boldsymbol{d}_j\right) + c_t L_i\, \delta_{ij}$$

$$\lambda^{\star} = \arg\min_{\lambda \ge 0} \tfrac{1}{2}\lambda^{\top} G\,\lambda - \delta\ell^{\top}\lambda$$

Tendon map: exactly linear, any bend

Pull-only: convex, unique, exact



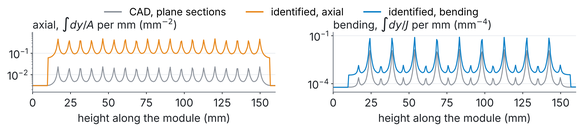
In a notch only two 4 mm bridges remain. The area falls eightfold and the bending stiffness about 280-fold, so the notches do the bending.

$\tau_{ic} \in \{0, 1\}$: tendon i passes slice c. $a_c = \int_c dy/A$, $C_c = \int_c dy/J$: slice compliances from the CAD. $c_t L_i$: tendon stretch.

B3 Identify the module law once, then verify

notch compliance	vs beam theory
axial	×20.2
bending, x and z	×5.9, ×5.3
torsion (twist test)	×2.45

solver check	difference
closed-form beam cases	< 1 µm
independent Cosserat rod	0.004 mm
automated tests	132 pass



Slice compliance along one module: the CAD, and the law identified from 675 FEM poses (SOFA) of one and two modules.

B4 Scientific ML: learn the law, not the solution

Learned law θ $C_\theta(y) = C_{CAD}(y)\, e^{\theta}$ 4 physical numbers → Exact physics pull-only QP + gravity → Shape ⟷ loss → Data FEM now, mocap next

fit θ through the solver

Next: a convex notch energy $\psi_\theta(N, \boldsymbol{M})$, with θ a function of geometry (Direction, step 2)

tip error vs FEM (mm)	seen	new mounting	4 modules
learn the **solution** (PINNs)	3 to 6	9 to 77	54 to 73, retrained
learn the law, in the solver	**0.2 to 0.4**	**0.2 to 0.4**	**0.50, same θ**

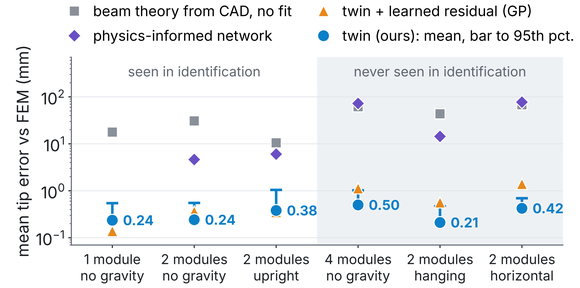
One θ for every stack and mounting. PINNs: forward and inverse, same data.

B5 Calibrate small, deploy large (FEM ground truth)

4 modules (640 mm) never used for identification — **mean tip error 0.50 mm**

2 modules, hanging mounting never used — **mean tip error 0.21 mm**

2 modules, horizontal mounting never used — **mean tip error 0.42 mm**



Dots: FEM centreline. Plot: mean and 95th percentile over 60 to 100 held-out poses.

On four modules, never used for identification, the mean tip error is 0.50 mm (0.08 % of length), against 62 mm for beam theory from the CAD alone. One pose takes 1 to 4 ms (250 to 970 Hz).

B6 Towards hardware: what the data can identify

problem (synthetic data)	remedy in the calibration
weight and stiffness confounded	weigh modules, fix the mass
equal pulls slacken distal tendons	pretension 1 : 1.75 : 2.25 : 2.5
0.4 mm moment-arm error	fit as 5th number, to 0.02 mm



Right: the parts designed for the four-module robot.

Conclusions

- **Design:** disc diameter sets curvature, length sets reach; nearly half the stroke is compression.
- **Model:** a module law learned inside exact physics transfers to unseen stacks and mountings.
- **Limit:** Part B is validated against FEM only; hardware is next.

Builds on: Camarillo et al., T-RO 2008 • Rucker & Webster, T-RO 2011 • Faure et al., SOFA, 2012 • Coevoet et al., SoftRobots, 2017 • Kim et al., arXiv:2507.10121 • Danesh et al., arXiv:2608.26273.

Direction A geometry-conditioned twin for inverse design

1 Hardware validation (now)
Motion capture, 4-module robot. Calibrate **5 numbers on 15 poses** of 2 modules; test 4 modules, upright and hanging.
Pass: tip error ≤ 2 % of length (13 mm), fixed before any data.

→

2 Law as a function of geometry
SWEEP generates designs; the twin identifies each: $\theta = f(D_{RD}, D_i, L)$.
Pass: beats CAD-only theory on designs held out entirely.

→

3 Inverse design
Target curvature + stroke budget → geometry, in milliseconds; print and verify.
If residuals demand: convex notch energy ψ_θ, friction, self-contact.

Outputs and my role

[1] D. Yadav, **O. M. Pawar**, Krishnapriya R, S. Gandhi, M. Vadali, "How Geometry Governs Compliance? A Parametric Study of Rolling-Disc Notched Tendon-Driven Continuum Robot," **under review, IEEE ICRA 2027.**
[2] O. M. Pawar et al., sensor-free digital twin of a modular notched TDCR, in preparation for IEEE RA-L, with an installable open package.

Role. Part A: SOFA 25.06 port, parallel sweeps, mesh and settling checks. Part B: model, solver, identification, calibration, baselines and software.

Report (PDF) Code (GitHub)